Shirpur Education Society's

# R. C. PATEL INSTITUTE OF TECHNOLOGY, SHIRPUR

**An Autonomous Institute**

( Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere )

**आर. सी. पटेल इन्स्टिट्यूट ऑफ टेक्नॉलॉजी, शिरपूर**

(स्वायत्त महाविद्यालय )

**Programme:** B.TECH (AIDS)/B.TECH (AIML) **Year: II/Semester IV (Exam Year: 2024-2025)**

**Subject:** Artificial Intelligence **Time:** 02:30 pm - 04:30 pm (02:00 Hrs.)

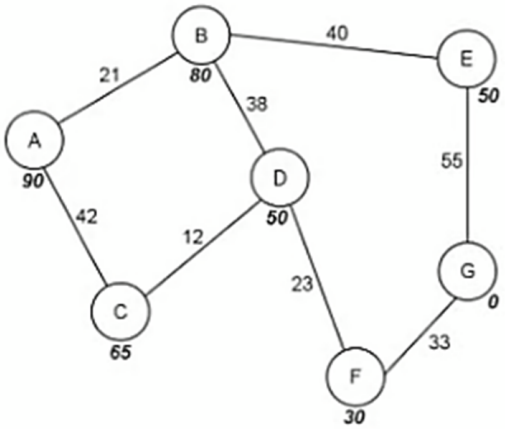
**Date:** 23 May 2025 **Max Marks: 60**

**END SEMESTER EXAMINATION EVEN SEM IV(2024-2025) REGULAR**

Instructions:
1. This question paper contains 6 pages
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. You can assume suitable data wherever necessary, state it clearly.

1. **15**

A. Implement 'A star algorithm' on the graph shown in following figure to find the shortest path and relevant cost. **10**




B. B. Solve any 1 **5**

i. Examine the two main strategies used in Bidirectional Search? **5**

**---------- OR ----------**

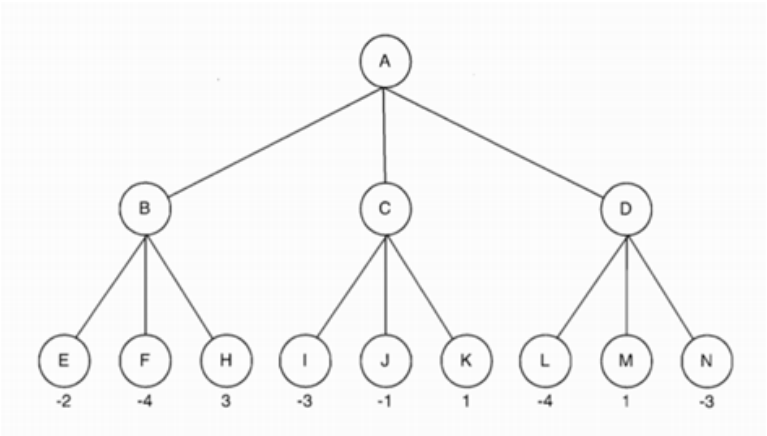
ii. Compare BFS and A* search . Also comment on the time and space complexity. **5**

2. **15**

A. Define heuristic function and explain its significance in informed search algorithms. **5**

B. B. **10**

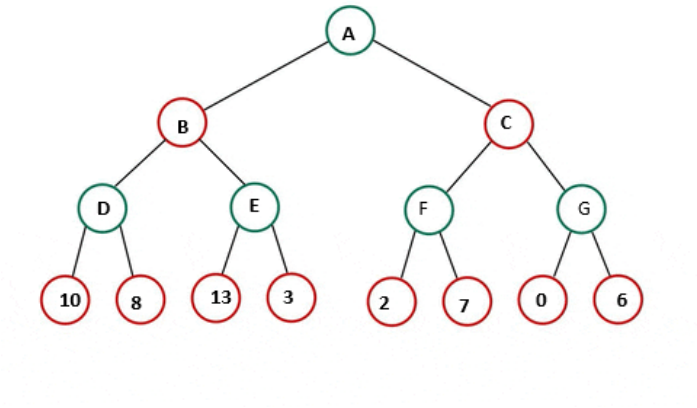
i. Find the best value for the Max Player using Minimax Procedure. Using Alpha Beta Pruning show the pruned branches: **10**




**---------- OR ----------**

ii. 2. Solve both a and b **10**

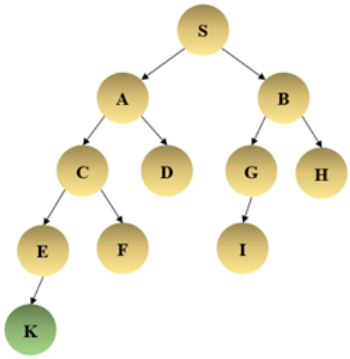
a. Find the value of root node using Minimax Algorithm **5**




b. Analyse the four main steps of Monte Carlo Tree Search and explain how each contributes to the balance between exploration and exploitation in decision making. **5**

3. **15**

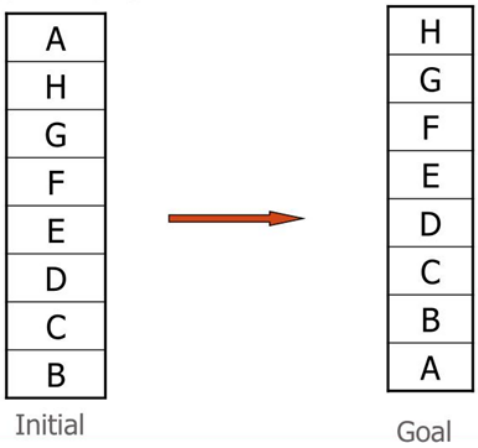
A. Given the following Graph. L =2 and apply DLS and comment if goal is achieved? Apply stepby step Iterative Deepening Search technique to reach the goal state. State the level at which goal is achieved. **8**




Note: Start State: S Goal State: K

B. B. **7**

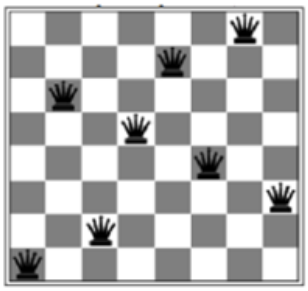
i. Consider the given blocks world problem with initial and Goal state. Assume the following two operations: Pick a block and put it on table, pick up a block and put it on another block. Solve the above problem using Hill climbing algorithm and a suitable heuristic function. Show the intermediate decisions and states. **7**




**---------- OR ----------**

For the 8-queens' problem, one configuration is as shown below: 7

ii.



a) In the context of computing the solution for this problem using the hill-climbing search method, we can define h = number of pairs of queens that are attacking each other, either directly or indirectly. For the configuration shown above, compute the value of h

b) We would like to use Genetic Algorithms to get to the optimal solution for the above problem. Propose a representation for the 8-queens problem that can be used to represent the above board position.

c) Take a random sample of 4 items from the population of possibilities for the 8-queens problem. Illustrate the following operators in the context of arriving at the optimal solution using Genetic Algorithms:

i. Selection

ii. Cross-over

iii. Mutation

4. **15**

A. Explain different issues in hill climbing also state the solution to come out of that problem. **5**

B. B. **10**

i. 1. a. **5**

1- Apply Forward chaining **5**

Following rule base:

R1: If it is raining and you have an umbrella, then go outside

R2: If it is not raining, then go outside

**Facts:** It is raining; you have an umbrella

**To Prove: -** To decide whether to go outside.

**---------- OR ----------**

b. Prove how you will apply resolution on given rules and facts **5**

goal: Patient has malaria

- Rules:
  - If patient has high fever and chills, then may have malaria
  - If patient has malaria, then prescribe anti-malarial drugs
- Facts: High fever, chills
- **Task:** justify drug prescription

ii. 2. **5**

a. 1- Explain what is a quantifier in first-order logic? Differentiate between the universal quantifier (∀) and the existential quantifier (∃). **5**

**---------- OR ----------**

b. 1- What are the advantages of hierarchical planning over classical planning? Provide examples of situations where hierarchical planning is more efficient. **5**